```
> ./pi-30 --benchmark
```

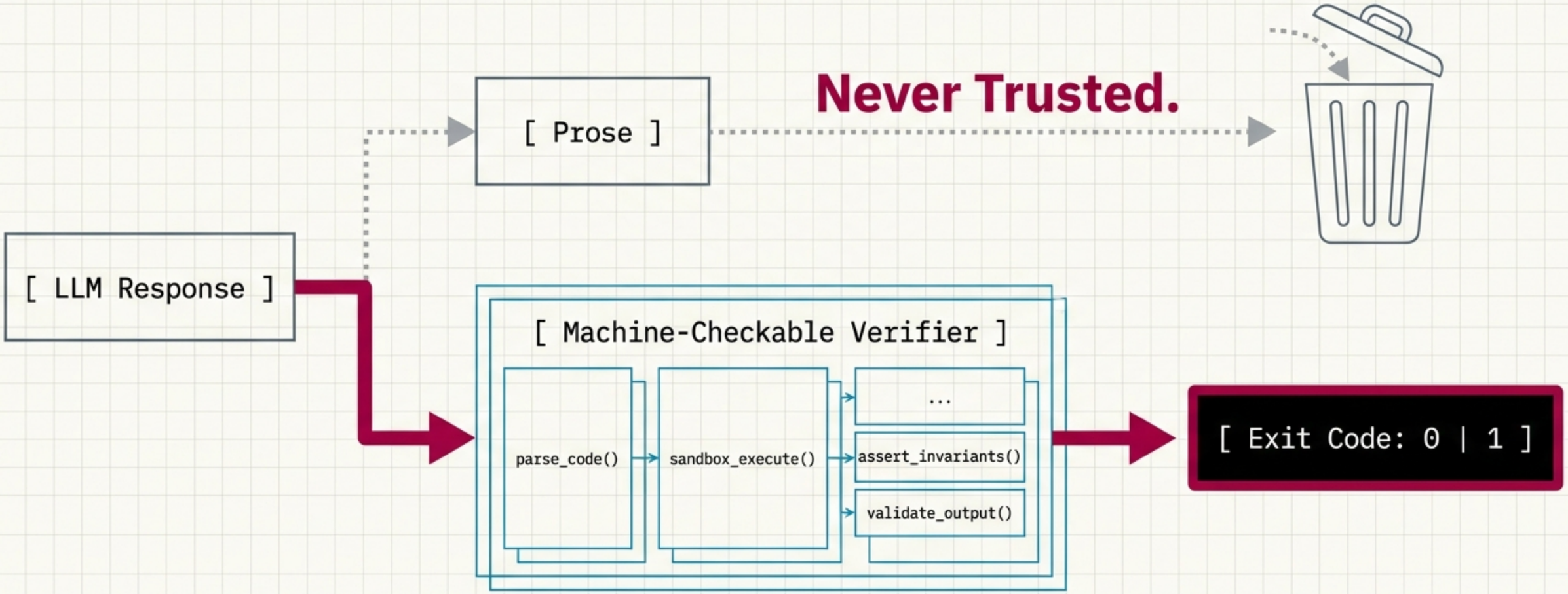
pi-30 Agent-Loop Fleet Benchmark

A 30-problem precision instrument for evaluating the Earendil Pi agent CLI.

NotebookLM

Exit codes are the sole source of truth.

Subjective prose parsing is deeply flawed.
In pi-30, the model's prose is completely ignored.
Scoring relies entirely on machine-verifiable execution.



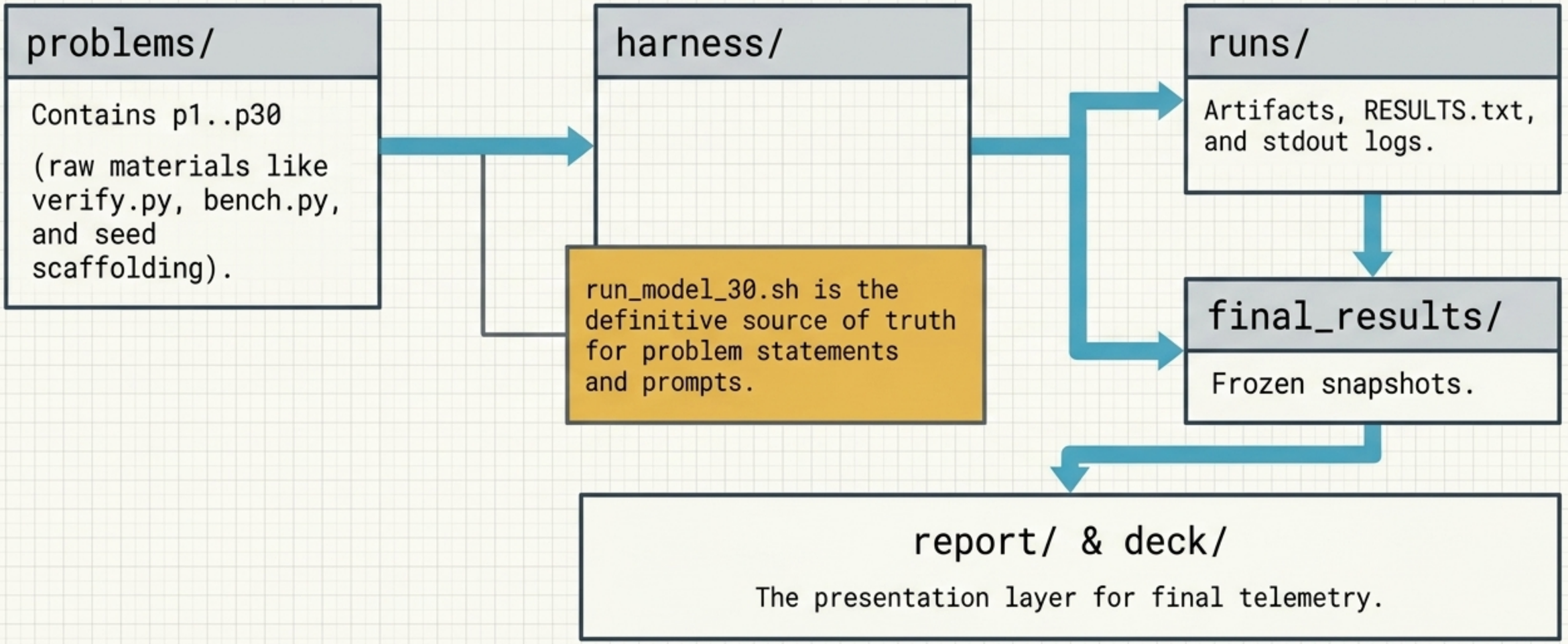
NotebookLM

Evaluating agents requires a multi-dimensional skills matrix.

5 Skills Matrix			
	**Write**	Build a utility from a spec	Tests I/O accuracy.
	**Fix**	Repair a seeded-broken module	Checks if test suite now passes.
	**Numerical**	Implement to a tolerance	Checks mathematical bounds.
	**Optimize**	Beat a performance baseline	Measures speed/efficiency thresholds.
	**Tournament**	Self-critique / converge	Checks evolutionary improvement via `converge_check.py`

NotebookLM

The repository serves as a self-contained test environment.



NotebookLM

Fleet orchestration relies on a standardized operational harness.

Enforces strict bounds on execution.

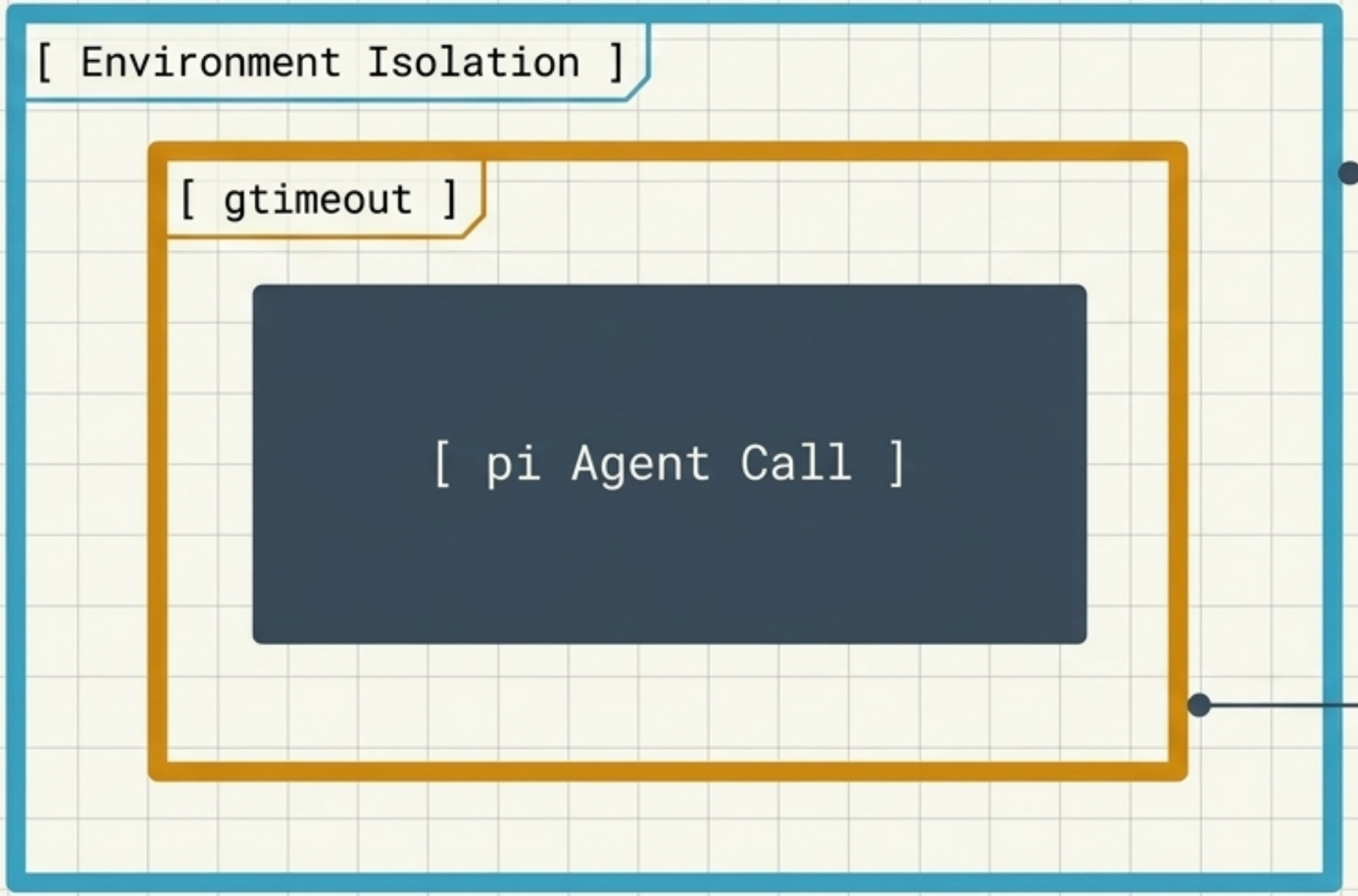
```
PI_TIMEOUT=360 bash run_model_30.sh
<provider> <model-id> <tag> > runs30_<tag>.log 2>&1
```

Integrates with CELS, Sophia, sparks, and OpenRouter (with **`429 backoff proxy`**).

Artifacts and **verdict exit codes** land directly in **isolated model lanes**.

NotebookLM

Engineering guardrails prevent infrastructure from corrupting results.



`unset PYTHONPATH` + `export CONDA_NO_PLUGINS=true` injected into `run_model_30.sh` to prevent leaked user-site packages from shadowing the real packages and causing false-fails in `pytest`.

Wraps every `pi` call and `score.py` execution to prevent pathologically hung endpoints from wedging wedging sequential lanes for hours.

NotebookLM

Immutable scoring rules ensure strict methodological rigor.

Single-Shot Canonical

True 30/30 requires a full k-of-N sweep.

Targeted reruns of missed problems are strictly forbidden.

Zero-Score Rule

Any `0/N` score is classified as an infrastructure stack failure (`BROKEN-<cause>`). debugged, never published as a capability failure.

Strict Segmentation

`Fleet-30` (P1-P30), `aggB-10`, and `mesh-extract-10` are isolated sets. They are never cross-merged to artificially inflate scores.

NotebookLM

Hardware nomenclature prioritizes transparency over internal codenames.

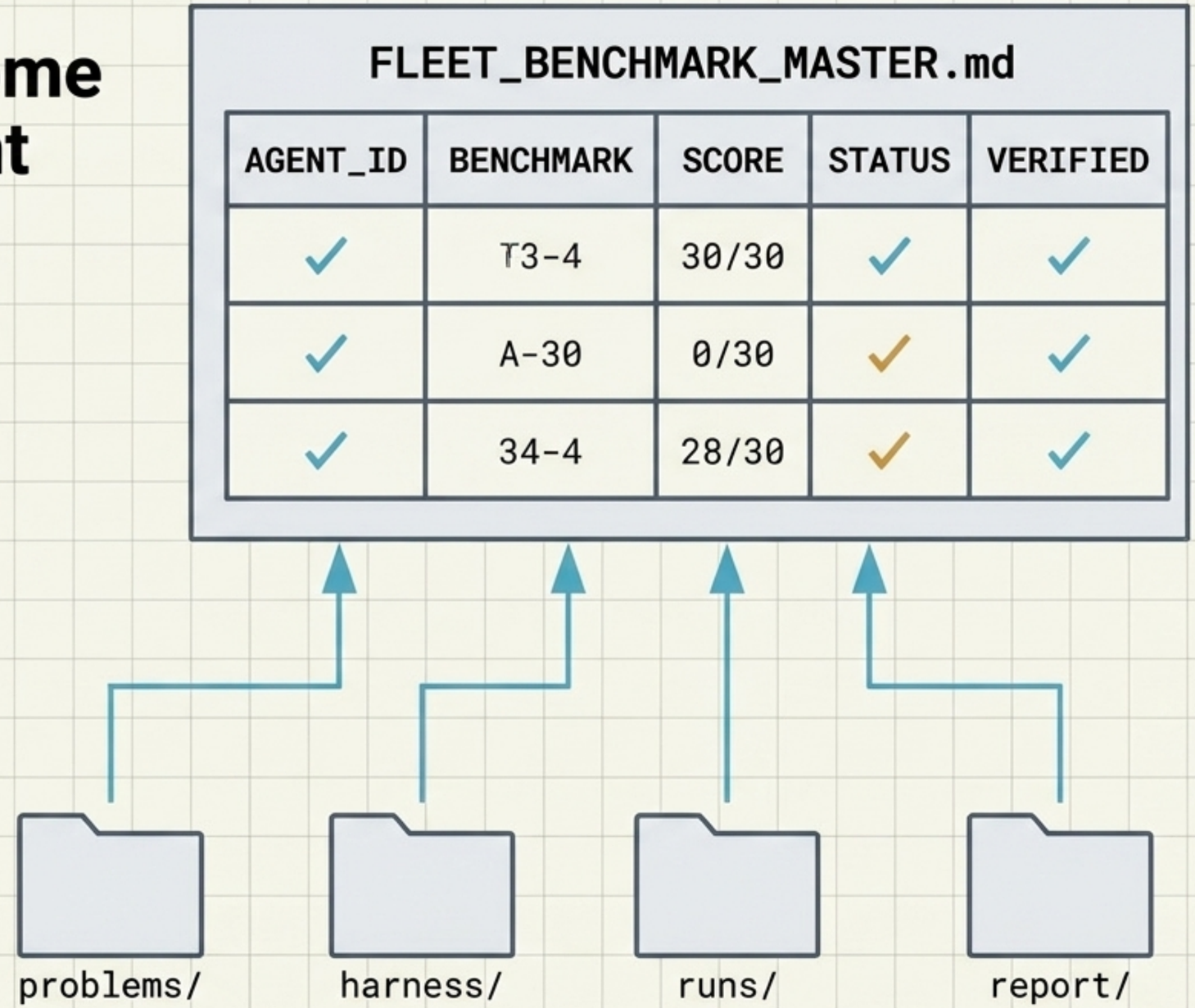
Backends are reported by their silicon and vendor reality, ensuring cross-platform reporting remains universally understood.

U1 `Intel PVC` →	(Data Center GPU Max 1550)
U2 `nVIDIA A100` →	(8xA100-80GB DGX)
U3 `local dgx` →	(Locally-served endpoints)
U4 `spark / OpenRouter` →	(ollama / External APIs)

NotebookLM

A single canonical home for reproducible agent evaluation.

From the problem seed to the final published report, `pi-30` provides the **benchmark**, the **execution harness**, the raw **evidence**, and the final **analytics** in one **immutable repository**.



NotebookLM